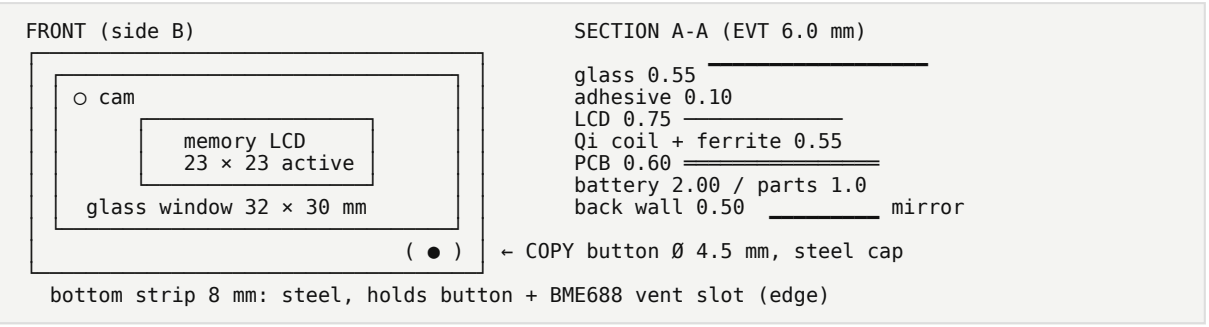
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LOT-01 — MECHANICAL: TWO-PART STAINLESS STEEL BODY
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DOCUMENT    HW-03 / MECH
ISSUE       2026.11.12 · REV A
MATERIAL    AISI 316L stainless steel (marine grade, low-nickel-release finish)
VENDOR      PCBWay CNC machining + polishing
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1 · OUTSIDE

Item	Value
Footprint	40.00 × 40.00 mm, corner radius R4.0
Height	5.0 mm target (DVT) · 6.0 mm EVT
Edge	0.3 mm × 45° chamfer on both faces
Back (side A)	Solid 316L, **mirror polish #8 (Ra ≤ 0.05 µm)**, no holes, no text (optional laser mark inside only)
Front (side B)	316L bezel plate with glass window + COPY button bore
Mass (est.)	38–45 g



2 · THE TWO PARTS

Part 1 — Back cup (mirror side). Machined from solid 316L. Floor 0.50 mm (EVT) / 0.40 mm (DVT), walls 1.2 mm. Four M1.2 threaded bosses in the corners (blind, do not break through the mirror face). Internal pocket for battery with 0.1 mm clearance. Polished **after** all machining; ships with protective film on.

Part 2 — Front bezel plate. 316L, 0.8 mm thick at the face, with:

- glass window 32 × 30 mm, 0.3 mm ledge for the cover glass;
- COPY button bore Ø 4.6 mm with a 0.2 mm step to retain the cap;
- vent slot 0.4 × 3 mm on the bottom edge (behind it: GORE vent → BME688);
- four counter-sunk holes for M1.2 Torx screws.

Seam. A 0.3 mm PEEK gasket between the two parts. It seals the joint and breaks the steel loop so 2.4 GHz energy can leave through the seam. Do not replace it with a metal shim.

3 · Z-STACK BUDGET

Layer	EVT (mm)	DVT (mm)	How DVT gets thinner
Cover glass	0.55	0.40	0.4 mm chemically strengthened glass
Optical adhesive	0.10	0.05	OCA film
Memory LCD	0.75	0.75	— (verify actual on datasheet)
Qi coil + ferrite	0.55	0.45	thinner ferrite sheet
PCB	0.60	0.40	4-layer 0.4 mm HDI

Layer	EVT (mm)	DVT (mm)	How DVT gets thinner
Battery (beside electronics strip)	2.00	1.60	1.6 mm cell, lower capacity (~ 65 mAh)
Back floor	0.50	0.40	thinner floor + rib
Tolerances / air	0.95	0.95	—
Total	**6.00**	**5.00**	

The camera module (≤ 3.0 mm) cannot sit on top of the PCB at 5.0 mm. It drops into a **cut-out through the PCB** and is held by its FPC, lens directly under a clear window in the glass ink mask.

4 · CNC + FINISH SPEC FOR PCBWAY

Field	Back cup	Front bezel
Material	SS 316L bar	SS 316L plate
General tolerance	ISO 2768-f	ISO 2768-f
Critical tolerance	Pocket depth ±0.02	Glass ledge ±0.02, flatness 0.03
Threads	M1.2 × 0.25, depth 2.0	—
Outside finish	Mirror polish #8	Bead blast fine **or** brushed (inventor's choice)
Inside finish	As machined	As machined
Marking	Laser inside: "LOT-01 · S/N"	none
Packing	Protective film, individual bag	individual bag
Inspection	CMM first article ×3, then 10 % sampling	same

Files to send: STEP (both parts), 2D PDF drawing with tolerances and finish callouts. CAD is the next deliverable (see roadmap, session 02).

5 · LOT DOCK (charger)

Ø 50 × 8 mm 316L polished ring, 1.0 mm ceramic or glass top, Qi TX coil directly under the top. The device lies **face-down** (glass to glass). A shallow 0.3 mm recess 40.4 × 40.4 mm in the top centers the device so the coils line up. USB-C on the side. Silicone ring on the base.

6 · RULES

1. No metal over the Qi coil, on either the device or the Dock.
2. No metal within 3 mm of the chip antenna except the PCB ground it needs.
3. The mirror back never gets a hole, logo cut or screw head.
4. Polish is the last machining step. Film stays on until packing.